

BIOINFORMATICS MINI-PROJECT 1

Project 1 — Structural Analysis of PDB 1FDL

Fab D1.3–Hen Egg-White Lysozyme Complex

Student End Report
Beginner-level research-profile project

1. Abstract

This first mini-project introduced structural bioinformatics through direct inspection of **PDB 1FDL**. The structure is an antibody–antigen complex containing the D1.3 Fab antibody and hen egg-white lysozyme. The work focused on learning how to navigate RCSB PDB and Mol*, identify chains, inspect the lysozyme component, and connect a 3D molecular view with biological meaning.

2. Project Question

What can be learned about an antibody–lysozyme complex by inspecting PDB 1FDL in a molecular viewer?

3. Structure Information

Property	Observation
PDB ID	1FDL
Type	Antibody–antigen complex
Method	X-ray diffraction
Resolution	2.50 Å
Chains	A: antibody light chain; B: antibody heavy chain; C: hen egg-white lysozyme
Lysozyme	129 residues; Gallus gallus

4. Methodology

I opened 1FDL in RCSB PDB, launched Mol*, inspected the molecule/chain controls, selected hen egg-white lysozyme (chain C; author chain Y), inspected its sequence, and viewed the complete complex in three dimensions.

5. Observations

A key beginner lesson was that the visible pieces in a molecular viewer should not simply be counted by eye as protein chains. The chain controls provide the reliable way to identify molecular components. The lysozyme sequence began with **KVFGRCELAA** and the database reports 129 residues.

6. Antibody–Antigen Interface

RCSB PDB describes lysozyme residues 18–27 and 117–125 as parts of a discontinuous antigenic determinant involved in antibody contacts. The interface includes hydrogen-bond and van der Waals interactions, with some contacts mediated by water. This shows how particular surface regions can participate in molecular recognition.

7. What I Learned

I learned how to move from a PDB record to a 3D molecular structure, identify chains, and connect sequence information with structure visualization. I also learned that antibody recognition depends on particular regions of an antigen.

8. Limitations and Future Work

No atomic contact counts, distance measurements, solvent-accessible surface calculations, docking, or RMSD calculations were performed. A useful next step would be to identify interface residues directly and quantify the interactions.

9. Conclusion

PDB 1FDL provided a strong first exercise in structural bioinformatics because it combines multiple chains, sequence inspection, molecular visualization, and a biologically meaningful antibody–antigen interaction.

10. References

RCSB Protein Data Bank, PDB 1FDL: <https://www.rcsb.org/structure/1FDL>

PDB DOI: <https://doi.org/10.2210/pdb1FDL/pdb>

Appendix — Project Screenshots

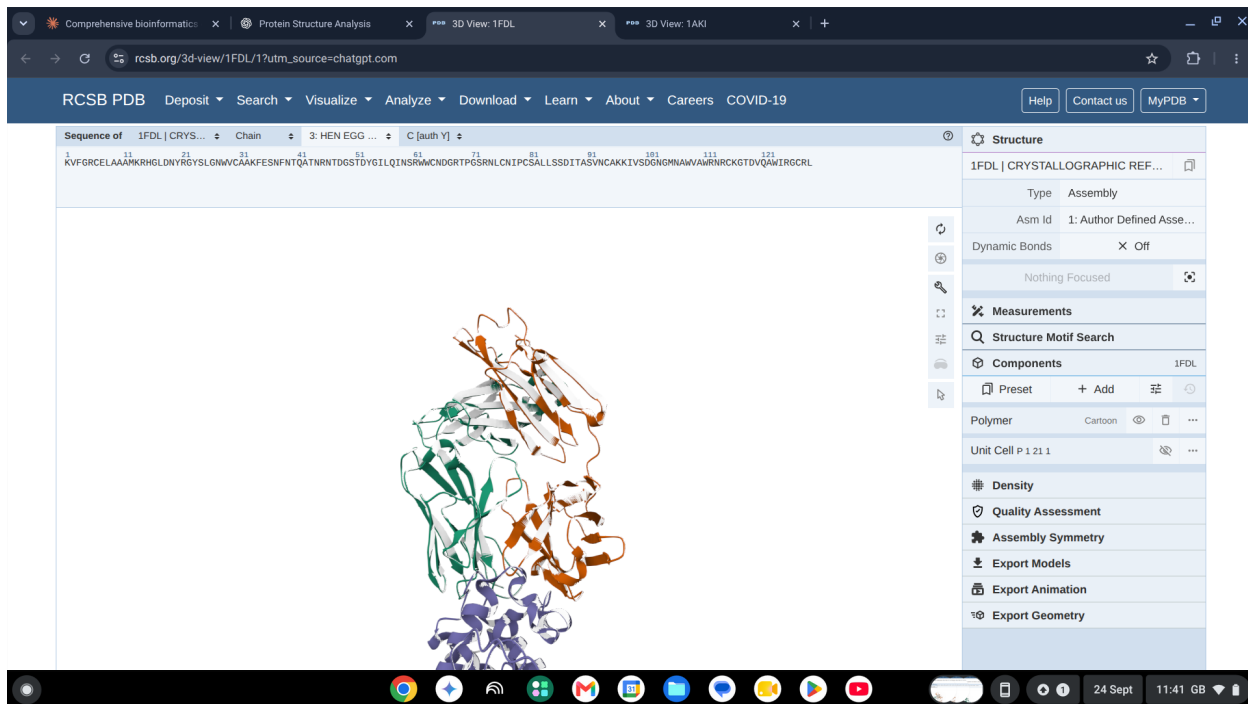


Figure A1. Mol* view of PDB 1FDL.

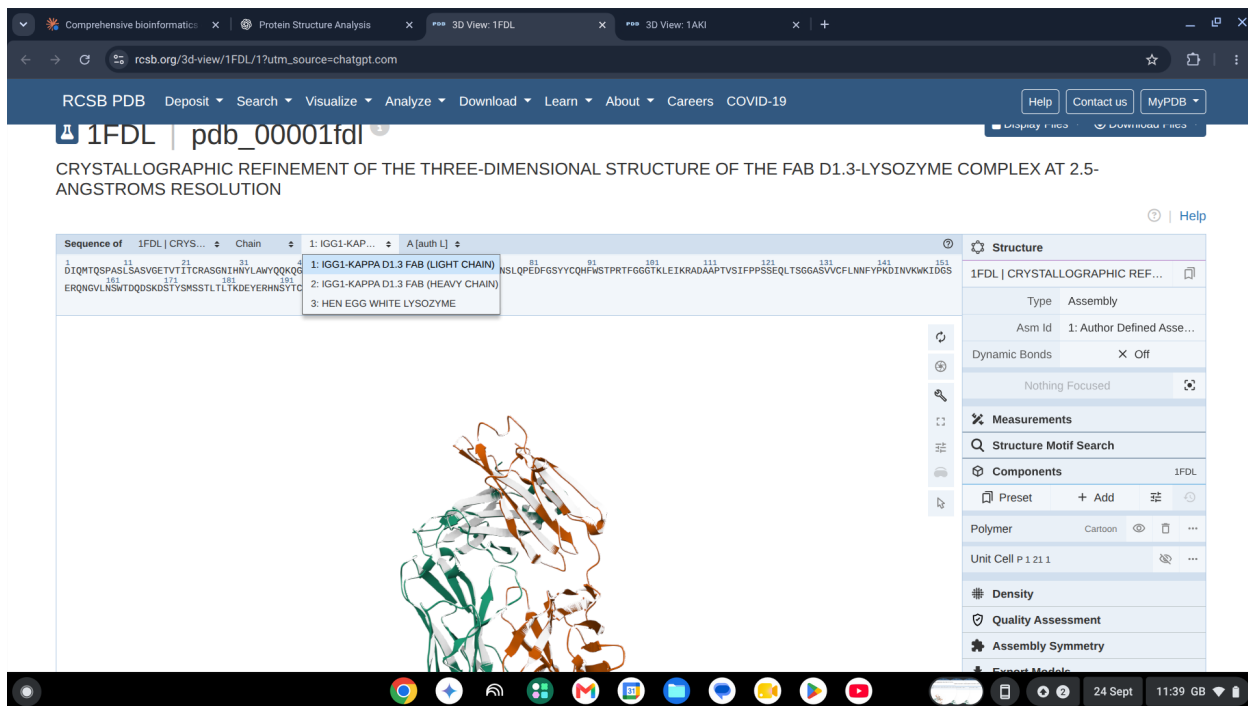


Figure A2. RCSB PDB / Mol* view of 1FDL.

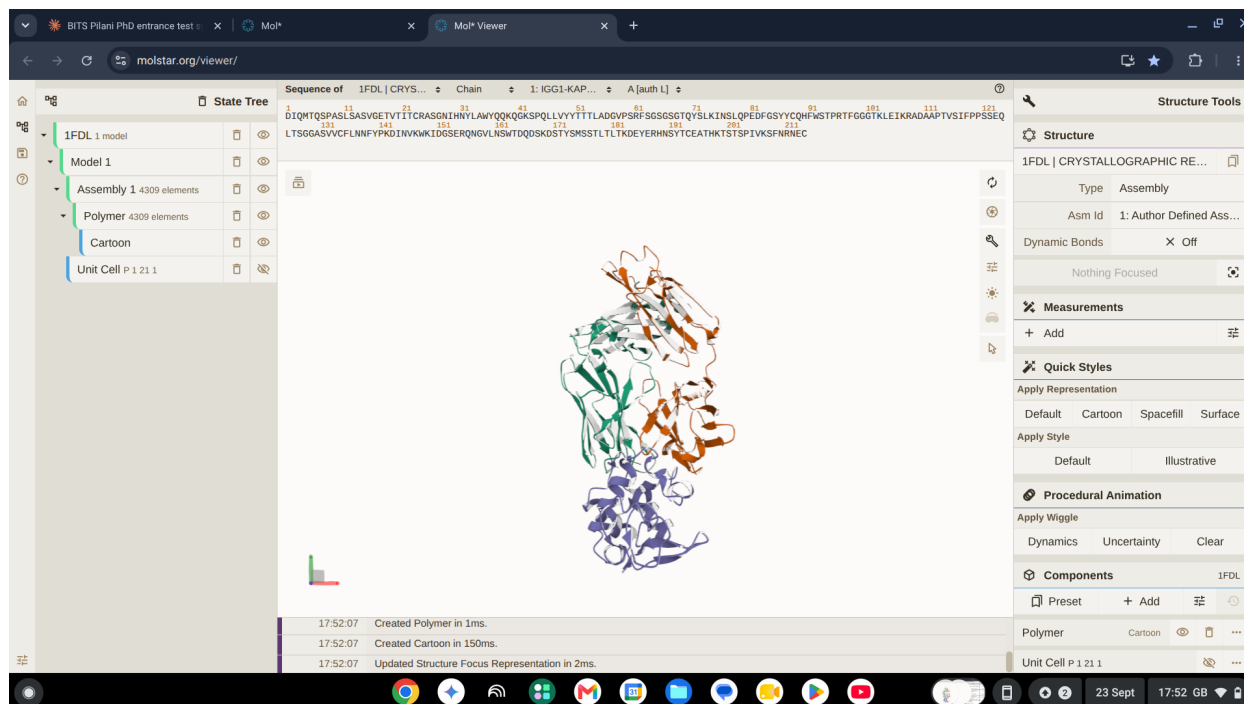


Figure A3. Additional recorded 1FDL structure view.